
Beyond Energies and Forces: OMol-Descriptors-4M, an Open Post-DFT Pipeline and Benchmark for Machine Learning in Chemistry

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Abstract

We present generator, an end-to-end computational chemistry post-analysis framework that processes repositories of molecular DFT wavefunctions and yields six partial-charge schemes (Hirshfeld, CM5, ADCH, Becke, Mulliken, Loewdin), four bond-order schemes, full QTAIM topology, fuzzy descriptors, and ORCA-derived globals. We built this by wrapping per-tier analysis routines around Multiwfn and ORCA’s orca_2mk1 utility, packaging the result into a scalable workflow for HPC clusters. Users can select compute tiers to trade fidelity for cost, then compact the resulting calculations into portable sharded Lightning Memory-Mapped Databases (LMDBs) keyed by job identity for downstream analysis and machine learning. To demonstrate the pipeline, we processed the recently-released OMol25 electronic structure dataset: a 4M-structure public subset of the 110M+-calculation 2025 release that ships the complete DFT outputs, including density matrices, energy gradients, ORCA .out files, and .gbw wavefunctions. By post-processing the wavefunctions, we extract atom-, bond-, and molecule-level descriptors across ~4M structures spanning 34 chemical verticals, supplementing the original dataset’s energy and force labels with physically-interpretable quantities. We name this dataset OMol-Descriptors-4M and present it as the first multi-vertical descriptor dataset at this scale beyond energies and forces. Analogous to the source dataset, we establish an evaluation protocol with explicit evaluative claims and scope boundaries: composition-ordered train/val/test splits via blake2b-hashed molecular formula, five domain-specific held-out stress tests (metal-ligand bond pairs, lanthanide-ligand bond pairs, reactivity, large systems, and high-charge regimes; ~60k structures total), and quantified per-descriptor noise floors derived from cross-method agreement on identical structures. Together with the open generation pipeline, these components let researchers reuse the protocol on new wavefunction repositories across molecular chemistry.

1 Introduction

Machine learning has fundamentally changed how we interpret, predict, and generate scientific information. Models now achieve expert or superhuman performance on language, vision, and biological structure tasks – most notably with AlphaFold2’s solution to the protein folding problem [Jumper et al., 2021]. In the chemical sciences, this shift has been equally transformative. Cheminformatics tools such as extended-connectivity fingerprints [Rogers and Hahn, 2010] coupled to tree-based models gave an early demonstration that learned descriptors could outperform expert intuition, as in the prediction of palladium-catalyzed reaction yields [Ahneman et al., 2018]. These approaches have since given way to continuous-filter graph neural networks [Schütt et al., 2018], directional

message-passing architectures [Gasteiger et al., 2020], and equivariant foundation models capable of stable atomistic simulation across nearly the entire periodic table [Batatia et al., 2025]. High-profile applications include the discovery of 2.2M stable inorganic crystals with GNoME [Merchant et al., 2023] and 3D-transformer pre-training on hundreds of millions of conformers for broad molecular property prediction [Lu et al., 2023].

A key enabler of this progress is the availability of large, well-curated quantum-chemical datasets. QM9 [Ramakrishnan et al., 2014] established the canonical small-molecule benchmark; SPICE [Eastman et al., 2023] and ANI-1x [Smith et al., 2020] extended coverage to biomolecular and reactive energy surfaces; OC20 [Chanussot et al., 2021] targeted heterogeneous catalysis; the Materials Project [Horton et al., 2025] and tmQM+ [Vargas et al., 2025] pushed into inorganic materials and transition-metal complexes. Alongside datasets, benchmarking suites such as MoleculeNet [Wu et al., 2018] and MOSES [Polykovskiy et al., 2020] have standardized evaluation across property prediction and molecular generation tasks. Yet across this landscape, datasets typically ship one descriptor family – energies and forces – over a narrow chemistry envelope. The recent OMol25 [TODO, 2025] is exceptional in scale ($>100\text{M}$ structures, 80+ elements, 34 chemical verticals) and releases wavefunctions, density matrices, and ORCA output files for 4M structures, making post-DFT descriptor extraction possible at scale for the first time.

We exploit this opening. By post-processing OMol25 wavefunctions through an end-to-end pipeline wrapping Multiwfn [Lu and Chen, 2012b] and ORCA [Neese et al., 2022], we extract six partial-charge schemes, three bond-order schemes, full QTAIM topology, fuzzy descriptors, and ORCA-derived globals across $\sim 4\text{M}$ structures. We name the result **OMol-Descriptors-4M** and contribute: (i) the dataset itself, (ii) the open generation pipeline, and (iii) an evaluation protocol with composition-ordered splits, five held-out stress-test suites, and quantified per-descriptor noise floors.

2 Related Work

Descriptor-rich quantum-chemistry datasets divide along two axes: the chemistry envelope they cover and the descriptor families they ship. Smaller, single-purpose releases such as QM9, QMugs, OE62, BDE-db, and tmQM+ [Ramakrishnan et al., 2014, Isert et al., 2022, Stuke et al., 2020, St. John et al., 2020, Vargas et al., 2025] sit below 10^6 structures and typically ship one descriptor family per dataset. Modern force-field training corpora – ANI-1x/2x, SPICE, GEOM, transition1x, and OMol25 itself [Smith et al., 2020, Devereux et al., 2020, Eastman et al., 2023, Axelrod and Gómez-Bombarelli, 2022, Schreiner et al., 2022, TODO, 2025] – routinely cross 10^6 – 10^8 structures but carry only energies, forces, and (in OMol25’s case) wavefunctions. We use those large geometry corpora as the structural source for OMol-Descriptors-4M, not as descriptor comparators.

2.1 Comparable-scale descriptor comparators

We selected four representative non-OMol25 datasets with comparable scale or a similar set of QM descriptors as direct comparators (Table 1). PCQM4Mv2 [Hu et al., 2021] reaches 3.4M drug-like organics drawn from PubChemQC [Nakata and Shimazaki, 2017] but ships exactly one descriptor, the HOMO-LUMO gap, at one level of theory. QM7-X [Hoja et al., 2021] reaches 4.2M conformers with a rich per-atom data inventory – Hirshfeld charges, dipoles, atomic C_6 coefficients, polarizabilities, van der Waals radii, and a PBE0+MBD energy decomposition – but only 6,950 unique compositions across CNOSCl with at most seven heavy atoms. Finally, QMugs [Isert et al., 2022] contains 2.0M conformers of 665k drug-like molecules with M06-2X/def2-SVP dipole and frontier-orbital energies. SchNet4AIM [Gallegos et al., 2024], though smaller with only $\sim 3.8\text{K}$ structures, is the nearest analog on the axis of dataset diversity. It contains only CHNO atoms with QTAIM charges, atomic localization indices (λ), and pairwise delocalization indices (δ).

Table 1: Comparator datasets used for the structural-overlap audit. “Compositions” counts unique molecular formulas. The bottom row is this work.

Dataset	Structures	Compositions	Chemistry envelope	Descriptors shipped
PCQM4Mv2	3.4M	3.4M	drug-like organics	HOMO-LUMO gap
QMugs	2.0M	665k	drug-like organics	DFT dipole, HOMO/LUMO/gap
QM7-X	4.2M	6.9k	CNOSCI, ≤ 7 heavy atoms	Hirshfeld; C_6 , α , vdW; PBE0+MBD
SchNet4AIM	3.9k	3.9k	CHON	QTAIM Q , λ , δ
Ours	4.0M	1.30M	TM, Ln, An, charged, reactive, protein-derived	6 charge, 3 bond-order schemes, full QTAIM, fuzzy, ORCA globals

2.2 Structural overlap

We confirm that the descriptor breadth above corresponds to genuine chemistry-space coverage by projecting OMol-Descriptors-4M against the four comparators in SOAP-UMAP space (Fig. 1). The embedding is fit on a balanced 500-molecule-per-source anchor (2,500 points total) using a 12-element common species set (H, B, C, N, O, F, Si, P, S, Cl, Br, I). The transition-metal, lanthanide, and reactive-geometry regions of OMol-Descriptors-4M sit well outside the support of every prior descriptor release: QM7-X collapses to a tight CNOSCI region, QMugs and PCQM4Mv2 occupy a drug-like manifold, and SchNet4AIM lands in a small CHON tip. No prior dataset is wide on both the chemistry-envelope and descriptor-family axes simultaneously, and none publishes a comparable coverage audit.

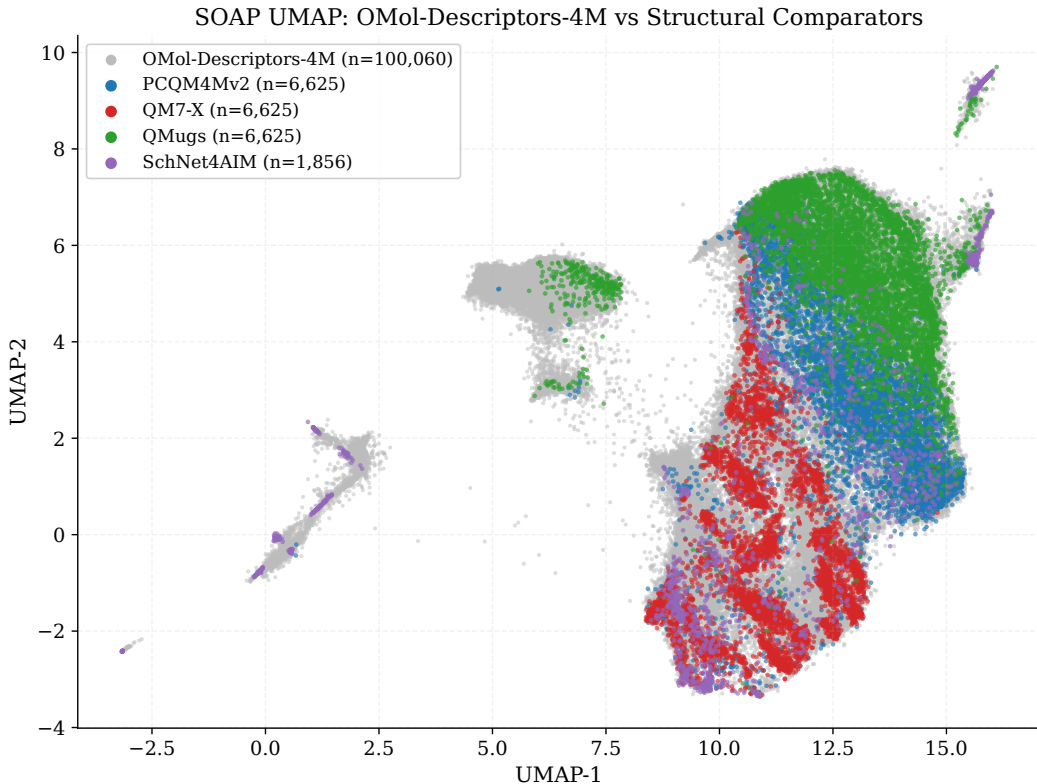


Figure 1: SOAP-UMAP embedding of OMol-Descriptors-4M against four structural comparator datasets. Each point is a molecule projected via UMAP fitted on a 500-molecule-per-source balanced subsample (2,500 anchor points total). SOAP descriptors use a 12-element species set (H, B, C, N, O, F, Si, P, S, Cl, Br, I) common to all five sources. OMol-Descriptors-4M (gray) is rendered as a background haze to expose the relative positions of PCQM4Mv2, QMugs, QM7-X, and SchNet4AIM.

3 Dataset

3.1 Descriptor Families

Partial charges. Six independent atomic partial-charge schemes are computed per structure. Hirshfeld [Hirshfeld, 1977], CM5 [Marenich et al., 2012], ADCH [Lu and Chen, 2012a], and Becke charges are produced by Multiwfn [Lu and Chen, 2012b] from the DFT electron density; Mulliken and Löwdin charges are extracted directly from the ORCA output [Neese et al., 2022]. Hirshfeld charges partition the electron density by the stockholder principle: each atom claims a share of the density at every point proportional to its free-atom promolecular density, yielding compact, chemically intuitive atomic populations. CM5 applies a pairwise connectivity-based correction to Hirshfeld charges to reproduce experimental dipole moments across diverse environments [Marenich et al., 2012]. Becke charges integrate the molecular density over smooth, fuzzy atomic basins defined by Becke’s numerical weight functions [Becke, 1988], providing a partition that is independent of the basis set and free of the basis-function artifacts that affect Mulliken and Löwdin populations. ADCH further corrects Hirshfeld charges for atomic dipole contributions to reduce basis-set dependence [Lu and Chen, 2012a].

Bond orders. Three schemes are universal across the corpus: Mayer [Mayer, 1983] and Löwdin bond orders from ORCA, and fuzzy bond orders from Multiwfn [Lu and Chen, 2012b, Becke, 1988]. Mayer bond orders derive from the density matrix and capture bond multiplicity beyond formal valence, including partial and dative character. Löwdin bond orders use a symmetrically orthogonalized basis, reducing the basis-set sensitivity that affects Mulliken-derived indices. Fuzzy bond orders integrate the exchange-correlation hole over Becke fuzzy-atom basins, providing a real-space, basis-set-independent measure of covalent bond character.

QTAIM topology. Critical points of the electron density are located via Multiwfn following Bader’s Quantum Theory of Atoms in Molecules [Bader, 1990]. Each bond critical point carries 26 scalar fields (ρ , $\nabla^2\rho$, ellipticity, kinetic/potential energy densities); ring and cage CPs, non-nuclear attractors, and atomic basin integrals are stored where present.

Fuzzy and surface descriptors. Per-atom Becke and Hirshfeld fuzzy electron-density integrations [Becke, 1988] are computed via Multiwfn for every structure, with spin-density counterparts for open-shell records. These provide atomic electron and spin populations that are independent of the one-electron basis and serve as a real-space cross-check against the partition-function-based charges in `charge.lmdb`. ALIE (Average Local Ionization Energy) surface properties – including surface volume, area, extrema, polarity index, and internal charge separation – are additionally extracted per molecule, encoding site-specific electrophilicity and reactivity complementary to the atomic-level descriptors.

ORCA globals. Total SCF energy, HOMO/LUMO orbital energies, molecular dipole, nuclear gradient, and SCF convergence metadata are parsed from the ORCA output file [Neese et al., 2022, Neese, 2012].

3.2 Storage Layout

Each split folder contains eight LMDB files keyed on a single string identifier $k = \text{relpath}(\text{folder}, \text{root}).\text{replace}("/", "_")$. Keys for each structure are shared across all lmdb files and they each contain different datatypes:

- `structure.lmdb` – pymatgen Molecule (positions, elements), total spin and charge, RDKit-derived bond list, and external identifiers. Entry point for any feature pipeline that needs geometry, formula, or atom counts.
- `charge.lmdb` – nested by partial-charge scheme. All seven schemes (`hirshfeld`, `cm5`, `adch`, `becke`, `mulliken_orca`, `loewdin_orca`, `mayer_orca`) ship per-atom charges; `hirshfeld` and `cm5` additionally ship a total dipole; `adch` and `becke` ship total dipole and per-atom atomic dipoles.
- `qtaim.lmdb` – per-structure dict of critical points keyed by atom ID for nuclear CPs ("0", "1", ...) or by atom-pair for bond CPs ("1_2"); 26 scalar fields per CP including ρ , $\nabla^2\rho$, ellipticity, kinetic and potential energy densities.

- `bond.lmdb` – nested by bond-order scheme (`fuzzy_bond`, `mayer_orca`, `loewdin_orca` universal across the corpus). Each scheme maps directed atom-pair strings of the form `"{i}_{El_i}_to_{j}_{El_j}"` to a scalar bond order.
- `fuzzy.lmdb` – per-atom Multiwfn fuzzy integrations keyed `"{i}_{El_i}"`. Universal entries are `becke_fuzzy_density` and `hirsh_fuzzy_density`; open-shell records additionally carry `becke_fuzzy_spin` and `hirsh_fuzzy_spin`.
- `other.lmdb` – molecular descriptors derived from ESP and ALIE surface analysis (volume, surface area, min/max, signed mean and variance, polarity index, internal charge separation) and four molecular polarity scalars (`mpp_full`, `sdp_full`, `mpp_heavy`, `sdp_heavy`).
- `orca.lmdb` – ORCA-derived globals: total energy, orbital energies, dipole, gradient, and SCF metadata. Filtered via `DEFAULT_ORCA_FILTER` when consumed by the converter.
- `timings.lmdb` – per-job step durations; provenance only, not consumed by the converters.

3.3 Data Quality and Per-Vertical Breakdown

Calculations and conversions to LMDBs are highly stable with five of the seven analytic LMDBs showing no corrupt or incomplete data storage. For `qtaim` and `orca` we have 25(0.0006%) and 1,607(0.0403%) failed calculations; the ORCA failures concentrate on physically hard cases (solvated biomolecules, scaled-separation electrolytes, open-shell TM redox), and geometry plus Multiwfn-side descriptors stay valid for those records. The failed QTAIM records where instances where the no BCPs were detected or where the Poincaré-Hopf formula was not obeyed. Cross-LMDB key alignment holds to within four records on a 4M-key index, so any descriptor combination joins on the full corpus. Each of the six charge schemes (Hirshfeld, ADCH, CM5, Becke, Mulliken, Löwdin), three bond-order schemes (fuzzy, Mayer, Löwdin), full QTAIM topology, and the standard ORCA globals reaches $\geq 99\%$ on every one of the 34 verticals, in other words, the same profile is available for a noble-gas dimer, a 200-atom solvated protein pocket, and a transition-metal redox complex. Full audit table, coverage heatmap, and per-vertical record counts: App. H and Table 3.

4 Pipeline

4.1 Architecture

The workflow pipeline is an orchestration layer over two external binaries, ORCA's `orca_2mk1` Neese [2012] and Multiwfn Lu and Chen [2012b]. The OMol25 release initially focused on energies and forces with a limited set of properties in their `aselmdb` metadata. They subsequently released the 4M electronic structure dataset hosted at Argonne National Lab with more complete coverage of electronic properties. The expanded 4M segment includes a slew of additional information including wavefunctions (as `gbws`), density matrices, and the raw outputs from the finalized ORCA calculations (`.out`, `.engrad`, `orca_property.txt` files, etc.). This 4M release is coupled to our post-processing pipeline to yield an even richer set of descriptors for the community to train models and evaluate beyond the standard gamut of energies and forces for machine-learned interatomic potentials (MLIPs).

Our pipeline reads in the `orca.out` and wavefunction file from each OMol job folder to commence. The `gbw`, in particular, is the foundation for all downstream analysis. As Multiwfn has more complete compatibility with `.wfn/.wfx` files, the first step the pipeline performs is a Molden conversion through `orca_2mk1` followed by EDF library loading inside Multiwfn so that ECPs are considered in downstream analyses. Without this EDF loading, there are potential inconsistencies with how ECPs are treated **TODO: cite multiwfn manual where it says that**. Folders that fail the EDF load are flagged automatically (see Sec. 3) and re-queued.

The user selects a calculation tier through a single integer, `full_set` $\in \{0, 1, 2\}$, which cascades through the four post-DFT families implemented in `qtaim_gen/source/data/multiwfn.py`. Tier 0 is the cheapest baseline with four partial charge schemes (Hirshfeld, ADCH, CM5, Becke), one bonding scheme (fuzzy bond order), full QTAIM analysis, and Becke/Hirshfeld fuzzy density with optional spin variants. Tier 1 adds three more charge schemes (VDD, MBIS, CHELPG), IBSI bond order, and ELF + MBIS fuzzy density. Tier 2 adds Bader (basin-integrated) charges, Laplacian bond orders, and Laplacian and gradient-norm fuzzy densities. Tier 2 enables the more expensive electrostatic potential geometry analysis (binned into our other group of calculations). The tiers are

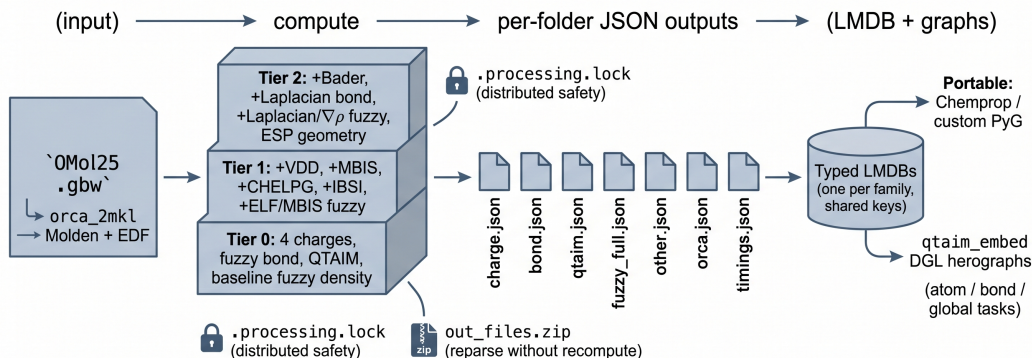


Figure 2: Overview of the qtaim_gen post-processing pipeline. Starting from an OMol25 .gbw wavefunction, the pipeline converts to Molden format (with EDF loading for ECP correctness), runs one of three calculation tiers through Multiwfn, and emits seven per-folder JSON outputs. These are converted to typed LMDBs sharing a common key, which can be consumed directly by graph-based frameworks or converted to DGL heterographs via qtaim_embed.

subsets of each other, this signifies that a user can "stack" calculations from 0 to 2 to incrementally add richer descriptors as they go. The tiered design is intended to give users a choice of how much compute they want to invest in the post-processing step, while allowing for future expansions. Table 2 summarizes the per-family contents of each tier; the QTAIM critical-point search is run at every tier and the ORCA-derived globals are always parsed.

Table 2: Per-family contents of each calculation tier (full_set). Tiers are strict supersets: tier k runs everything in tiers $0, \dots, k$. Schemes shown in a column are those added at that tier. Source: qtaim_gen/source/data/multiwfn.py.

Family	Tier 0 (baseline)	Tier 1 (+adds)	Tier 2 (+adds)
Partial charges	Hirshfeld, ADCH, CM5, Becke	VDD, MBIS, CHELPG	Bader (basin-integrated)
Bond orders	Fuzzy bond order	IBSI bond order	Laplacian bond order
Fuzzy density	Becke density, Hirshfeld density, Becke spin, Hirshfeld spin	ELF, MBIS density, MBIS spin	Laplacian ρ , gradient-norm $ \nabla \rho $
Other (scalar fields)	Geometry (ellipticity), ALIE	—	ESP geometry

Always-on (run at every tier): full QTAIM critical-point search (atoms, bond CPs, ring CPs, cage CPs; ρ , $\nabla^2 \rho$, ellipticity, energy density); ORCA .out parsed globals (energies, orbital eigenvalues, dipole, gradient, Mulliken / Loewdin charges, SCF metadata, quality flags). Note: ORCA’s Mayer population analysis block reports its QA column as “Mulliken gross atomic charge”; this is stored for completeness but is not treated as an independent charge scheme in downstream analysis.

Every job folder carries its own state of completed calculations and whether it is being worked on. A .processing.lock file with a stale-mtime threshold protects against conflicting work across distributed workers, and a per-folder out_files.zip archives every Multiwfn .out stream so a folder can be reparsed or analyzed without re-running conversions or Multiwfn. Validation is structural rather than numeric: each tier defines the JSON keys that must be present, and the pre-validation pass

203 treats a folder as done only if all expected keys are populated. ECP-failed folders are detected by
204 scanning the archived ADCH/CM5 logs for the Loading EDF library finished banner.

205 The per-folder outputs are seven JSON files: `charge.json`, `bond.json`, `qtai.json`,
206 `fuzzy_full.json`, `other.json`, `orca.json`, and `timings.json`. The first five are Multiwfn-
207 derived and indexed by method name. The full per-method schema for each tier is enu-
208 merated in Sec. 3; we only sketch the families here. `orca.json` is parsed directly from
209 the ORCA `.out` file (energies, orbital eigenvalues, dipole, gradient, SCF metadata, Mul-
210 liken/Loewdin charges, and quality-filter fields) by a single-pass enum state machine implemented
211 in `qtai_gen/source/core/parse_orca.py`. We chose this over a general parser like `cclib` ?
212 because the state machine streams the file once, has zero runtime dependencies beyond the standard
213 library, and only extracts the fields we need. `timings.json` stores per-step wallclock seconds and is
214 carried through to LMDB as a provenance record so every downstream artifact can be audited back
215 to its compute cost. We do note that `timings.json` does not store number of cores, architectures, or
216 other details for exact comparisons but is still useful for relative analysis.

217 Once the desired tier is complete, the JSON outputs are converted into a set of typed LMDB files
218 via `json-to-lmdb`: one LMDB per descriptor family (geometry, charge, bond, qtai, fuzzy, other,
219 orca, timings) sharing a common key derived from the relative path of the job folder. The compact
220 LMDB representation is portable and self-describing, so it is directly usable by external graph
221 models such as Chemprop **TODO: cite** or any custom PyG/DGL pipeline. For users of our companion
222 package `qtai_embed`, the `generator-to-embed` entrypoint composes the typed LMDBs into PyG
223 heterographs through the `GeneralConverter` class, with configurable bonding schemes (structural
224 cutoff, fuzzy, IBSI, or QTAI), charge filter, fuzzy filter, and ORCA-globals filter. A second
225 QTAIConverter class is retained specifically for QTAI-only graphs, where bond edges are
226 defined by QTAI bond critical points rather than by structural cutoffs. The resulting graphs expose
227 atom-level, bond-level, and global-level targets simultaneously.

228 **Features flagged for trimming before camera-ready.** While auditing the codebase
229 to write this section, two threads of dead or near-dead capability surfaced. First,
230 `qtai_gen/source/core/parse_critic.py` (Critic2 critical-point parser) is no longer imported
231 by any active workflow; the only remaining reference is an optional `critic2_cmd` field in
232 `create_files.py` that gates a legacy reaction workflow. We intend to drop the Critic2 depen-
233 dency entirely so that Multiwfn is the sole QTAI backend. Second, the older QTAI-only job
234 runners under `scripts/old/` (`generator_run.py` and the QTAI-specific Parsl runner variants)
235 predate the multi-method full-runner family and can be removed without loss of capability. We retain
236 QTAIConverter itself because it underpins the bond-classification task.

237 4.2 Sharded Execution and LMDB Merge

238 LMDB is single-writer and exhibits lock contention under naive parallel writes, so `json-to-lmdb`
239 does not attempt to share a single output environment across workers. Instead, each parallel job is
240 given a shard index $i \in \{0, \dots, N - 1\}$ and a total shard count N , and partitions the input folder list
241 by index modulo N . For the flat `root/category/subset/job` hierarchy this is a glob plus a modulo filter;
242 for jagged hierarchies such as OMol4M, where job folders sit at variable depth, the user supplies an
243 explicit `-folder_list` file of absolute paths and the same modulo partitioning applies to the line
244 index. In both cases the LMDB key is `relpath(folder, root_dir).replace(os.sep, "__")`,
245 which is identical across every descriptor family, so the per-family shards always share keys and
246 joining is trivial.

247 Each shard writes into its own subdirectory as `shard_{i}.lmdb`. When the last shard finishes its
248 work it drops a sentinel file (`shard_N.done`); the merge step waits on the full sentinel set and then
249 concatenates per-family shards into a single consolidated LMDB per family. The same shard/merge
250 pattern is implemented at the converter stage in `converter.py`, so graph construction can also be
251 parallelized and joined into one final per-vertical graph LMDB.

252 4.3 HPC Portability

253 We exercised the pipeline on three production HPC centers, each with a different sched-
254 uler and a different parallelism paradigm. ALCF Polaris uses PBS Pro and a controller-and-

workers Parsl topology: one login or batch node hosts the Parsl controller, which farms individual job folders out to PBS-allocated worker nodes through the PBSProProvider executor in `qtaim_gen/source/utils/parsl_configs.py`. NERSC Perlmutter uses Slurm and the same topology with the SlurmProvider executor; the only meaningful change is the queue name and the per-node thread budget. Both centers scaled cleanly because the controller node sees only the work queue and the worker nodes see only their assigned folders.

LLNL Tuolumne uses Flux as its native scheduler, and we did not get the controller-and-workers topology to behave reliably under Flux. The fallback used on Tuolumne was the single-node paradigm: a Slurm or Flux allocation grants a node, and a local Parsl ThreadPoolExecutor chews through the local job queue with no remote dispatch. This is the same path used for laptop and workstation runs (full-runner), parameterized by `n_workers` and `n_threads_per_job`. NERSC and ALCF can also be run in this single-node mode if a user prefers not to manage a controller, at the cost of losing multi-node scaling.

The submission scripts and Parsl configs for all three centers are checked in under `qtaim_gen/source/utils/parsl_configs.py`; PBS, Slurm, and the single-node Flux fallback are all minimal edits of the same template. Reproducing a run on a new center is expected to require swapping the provider class and the queue/account fields, nothing more.

4.4 Reproducibility

The pipeline pins no specific version of ORCA or Multiwfn at runtime; some compatibility across versions of either is expected though we have not exhaustively tested versions of each. For example, we also tested Orca5Neese et al. [2022] wfn conversion where our code handles its treatment of ECPs and as well as Orca 6.0 and 6.1. In the case of Multiwfn, any version that follows the same menu structure should work though we only tested 3.9 and 3.8. The dataset reported here was produced with ORCA 6.0.0, the same release used for OMol25 and Multiwfn 3.8. In our experience, the `orca_2mkl` version should match that of the gbws from input DFT calculations.

Determinism in the pipeline is structural rather than numeric: the LMDB key derivation, the per-tier set of expected JSON keys, and the parsing logic for both Multiwfn and ORCA outputs are fixed by code and covered by the test suite under `tests/`. Numeric reproducibility of the underlying QM calculation is the responsibility of ORCA itself and is not something we modify.

The test suite serves as the end-to-end smoke test: `tests/test_lmdb.py` and `tests/test_general_converter_filters.py` together exercise json-to-lmdb, sharded conversion, merge, the GeneralConverter graph construction, and scaler fit/apply on a small fixture set under `tests/test_files/lmdb_tests/`. Running `pytest -q` on a fresh checkout is therefore a ground-truth check that a user’s local toolchain produces the same LMDB and graph artifacts we ship.

5 Cross-Method Noise Analysis

5.1 Methodology

5.2 Charge-Method Comparison

5.3 Per-Vertical Noise Floors

6 Census

OMol-Descriptors-4M comprises ~ 4 million structures across 34 chemical verticals, indexing ~ 1.3 M unique compositions. At the descriptor level the combined corpus carries ~ 218 M atom-level records, over 30 million charge-scheme records (per-job, per-method), ~ 1.83 billion bond-scheme records. All counts in this section are corpus-level totals prior to the train/val/test partition.

6.1 Bond records per vertical

Figure 3 shows the per-vertical bond record count, decomposed by scheme. Four bond-order schemes are reported on every structure: Mayer (ORCA), Löwdin (ORCA), QTAIM, and fuzzy bond order (Multiwfn).

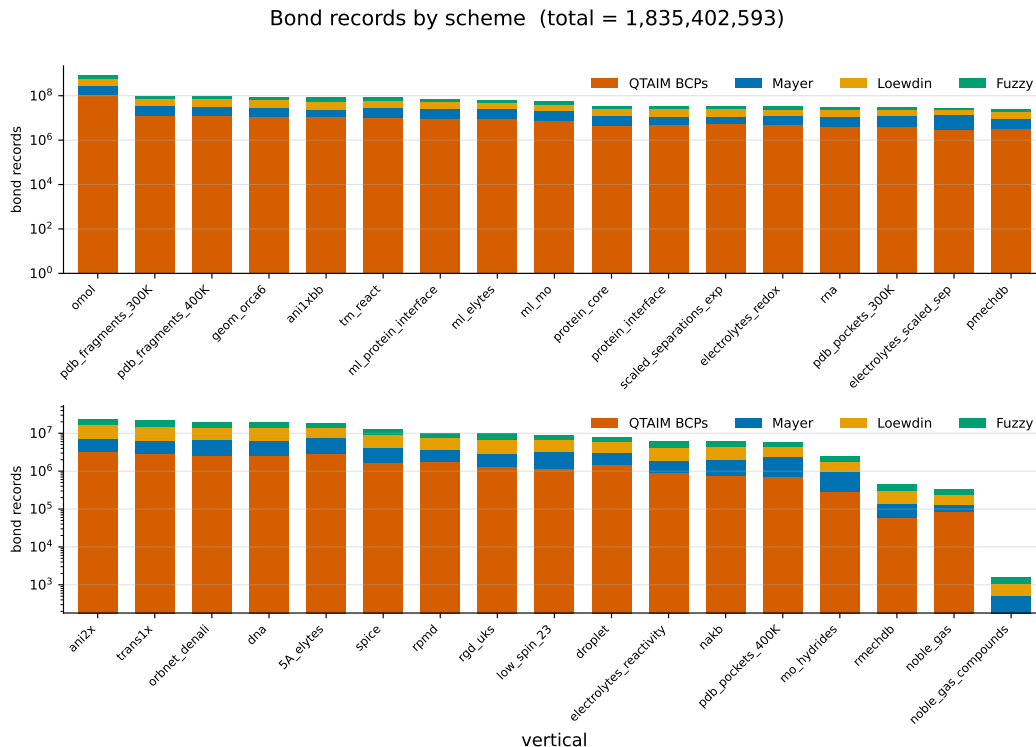


Figure 3: Per-vertical bond-order record counts, stacked by scheme (Mayer, Löwdin, fuzzy) on a log axis. Verticals are sorted by total bond-record count. The three schemes report on the same set of bonds, so counts represent records, not unique bond instances. Corpus total across the three schemes is 1.58 billion records.

6.2 Charge-scheme records per vertical

Charge-scheme records track how many partial-charge methods executed successfully on each structure. The corpus carries ~ 30 M datapoints. The full per-vertical breakdown is given in Table 3.

7 Evaluation Protocol

7.1 Tasks

- T1 – Per-atom partial charge regression.** Multi-target: ADCH, Lowdin, Mulliken, Hirshfeld, CM5 jointly per atom. Headline: per-method, per-element MAE and Pearson r on the test split. Long-tail per-element table in appendix.
- T2 – BCP property regression.** Targets: ρ , $\nabla^2 \rho$, ellipticity at every QTAIM bond critical point. Headline: per-element-pair MAE on ρ , Pearson r per pair, stratified by bond type (single, double, aromatic, metal-ligand) inferred from QTAIM topology.
- T3 – Bond classification.** Binary: does a QTAIM BCP exist between an atom pair within $2 \times$ sum-of-covalent-radii. Compared against Mayer-ORCA / Löwdin-ORCA / fuzzy agreement. Macro-F1 plus per-element-pair confusion stats.

- T3 sharpening (open): mine the disagreement subset (QTAIM-bonded vs distance-bonded) once `bond.lmdb` and `qtaim.lmdb` are populated end-to-end; that subset is the discriminating slice.

7.2 Main Split

- Composition-ordered, deterministic: each `formula_hill` hashed via BLAKE2b Aumasson et al. [2013] into $N = 10$ buckets.
- Buckets 0-7 train, bucket 8 val, bucket 9 test. Holdouts (§7.3) are excluded before bucket assignment.
- Realized post-holdout sizes: train 3,189,390 / val 399,690 / test 415,913.
- Bucket-size variance under 5% of the mean; per-bucket unique-formula count $\approx 112k$ each, confirming the partition balances composition and not row count.
- Implementation: `split_descriptor_lmdbs.py` runs per-vertical, sharded over verticals, race-free under parallel `xargs`; per-vertical `split_assignment.json` is the durable record.
- Verticals are not sampled uniformly because the split is global by formula. A per-vertical post-split distribution table will be published in the appendix as a transparency artifact.

7.3 Held-Out Evaluation Sets

The held-out sets are constructed independently of the main split and materialized before any model is trained. Membership is decided from the manifest, and from `bond.lmdb` for the two metal-ligand sets, then removed from the train, val, and test buckets so the same structure is never simultaneously evaluated as in-distribution test and as a stress test. Overlaps between held-out sets are allowed (a structure can be both H7-large and H8-high-charge), but each set is materialized once and reported separately so the reader can attribute error to the axis being stressed. None of these sets is used for hyperparameter selection or early stopping; numbers are reported once at the end of training.

The headline generalization signal per set is the gap $\Delta = \text{metric}_{\text{held-out}} - \text{metric}_{\text{main test}}$, reported alongside the absolute number so a reviewer can read the floor (test) and the stress response (Δ) on the same line. A small Δ on H7 means size extrapolates; a large Δ on H1 means TM-ligand chemistry is being memorized at the pair level.

Three of the five sets (H1, H6, H7) inherit the hardest evaluation axes from the original OMol25 paper directly: rare transition-metal-ligand bond pairs, rare lanthanide-ligand bond pairs, and large-system extrapolation. Where OMol25 publishes a hold-out construction, we lift it unchanged onto our manifest rather than redesigning it, both for comparability and because the OMol25 axes were already chosen to stress chemically meaningful failure modes.

Sets and sizes. The five held-out sets together cover roughly 60,719 structures ($\sim 1.5\%$ of the dataset; overlaps allowed):

- **H1 – Metal-ligand pairs (15,030 structures).** Eleven (TM, partner) bond pairs sampled stratified across log-spaced frequency bands with shape (3, 2, 2, 4, 0) and `seed=87`. Selection is driven by *true* bonds (Mayer / fuzzy bond order ≥ 0.3) extracted from `bond.lmdb`, not element co-occurrence in the formula; the full procedure is in Appendix D.1. Tests whether the model can predict descriptors for transition-metal bond chemistry it never saw during training.
- **H3 – Reactivity (12,507 structures).** Composition-stratified subsample of three reactivity-flavored verticals: `tm_react` (5k), `electrolytes_reactivity` (5k), `pmechdb` (2.5k). Per-vertical formulas are chosen by deterministic BLAKE2b hash so a reader can reproduce membership without rerunning the sampler. Tests transfer to non-equilibrium and transition-state-adjacent geometries.
- **H6 – Lanthanide-ligand pairs (2,589 structures).** Same construction as H1 applied to (Ln, partner) bonds; shape (3, 2, 2, 1, 0), `seed=190`; full procedure in Appendix D.2. The set is smaller than H1 because the Ln-bonded subset is itself smaller (66,677 structures vs 651,742) and the highest-frequency band is empty.

- **H7 – Large systems (18,200 structures).** Strict cutoff `n_atoms > 250`. Inherits OMol25’s large-system stress axis unchanged. Targets size extrapolation in the regime where several descriptor methods (notably fuzzy integration and grid-based QTAIM) are most expensive, and where the per-atom signal-to-noise ratio in DFT outputs is hardest to maintain.
- **H8 – Large net charges (12,393 structures).** Strict cutoff `net_charge_abs > 4` (equivalently $|q| \geq 5$). Targets the high-charge tail where partial-charge schemes disagree the most (cf. §5) and where SCF behavior is most sensitive to functional and basis choice.

We deliberately use pair-level held-outs (H1, H6) rather than whole-element held-outs because the goal of this benchmark is to produce models that are usable across the periodic table, not models that have a hole where transition metals or lanthanides should be. Dropping every `has_tm` structure would remove 17.24% of the dataset and evict whole verticals (`low_spin_23`, `tm_react`, `ml_mo`) from training entirely; dropping every `has_lanthanide` structure would remove 1.78% and similarly empty the Ln-bearing verticals. In both cases the resulting model would simply have no signal to predict on for those elements at inference time. Holding out specific (metal, ligand) pairs keeps every element represented in training while still measuring rare-chemistry transfer, which is the construction the OMol25 paper uses for the same reason.

7.4 Holdout-Aware Splitting Pipeline

- **Step 1 – Pull holdouts.** `pull_holdout_records.py` reads each holdout CSV, materializes per-set LMDBs under `data/holdouts/<set>/`, and emits a `manifest_holdout.parquet` keyed on the LMDb key convention `relpath(folder, root_dir).replace(os.sep, "__")`.
- **Step 2 – Exclude.** `multi_vertical.load_exclusion_set` re-derives LMDb keys from the manifest `rel_path` column and unions H1/H3/H6/H7/H8 into one per-vertical exclusion set.
- **Step 3 – Split.** `split_descriptor_lmdbs.py` reads `structure.lmdb`, drops excluded keys, assigns each remaining key to train/val/test by `BLAKE2b(formula_hill) % 10`, then partitions each of the 8 descriptor LMDBs (`structure`, `charge`, `qtaim`, `bond`, `fuzzy`, `other`, `orca`, `timings`) into the assigned split.
- Output layout: `<output_dir>/<vertical>/<split>/<descriptor>.lmdb`.
- HPC parallelism: `-include_verticals X + -merge_from DIR`, mirroring the pull pipeline. Per-vertical `split_assignment.json` is the durable, race-free record (no global plan file).
- Lock-step audit: across all 8 descriptors, `source row count = train + val + test + excluded` for every vertical (verified on `ani2x`: 215,642 source = 168,783 + 17,417 + 29,442, $n_{\text{excluded}} = 0$).

7.5 Metrics

- Per task: MAE, RMSE, Pearson r .
- **T1** stratifications: per-method, per-element, per-vertical, per-charge-bucket.
- **T2** stratifications: per-element-pair, per-bond-type (covalent / metal-ligand / H-bond), per-vertical.
- **T3** stratifications: macro-F1, per-element-pair confusion, per-distance-bucket recall.
- Held-out sets: same metrics, plus the gap to the corresponding main-test number.

7.6 Methodology and Reporting

- Train on the 8 train buckets. Tune on val (bucket 8). Final numbers reported once on test (bucket 9) and on each held-out set, with no further tuning.
- Held-out sets must not be used for hyperparameter selection or early stopping.

- Reference score-predictions CLI deferred to v2: shipping it for v1 without a calibrated reference model would expose the protocol to garbage-in adoption. v2 will release the script alongside a reference model whose error sits between the cross-method noise floor and naive baselines.
- Cross-method noise floors from §5 are a lower bound on achievable error: any model below the floor is overfitting noise, not learning chemistry. Future work using this benchmark should report the floor alongside model error.
- Per-vertical post-split distribution: a transparency artifact (notebook + appendix table) listing realized train/val/test counts per vertical so readers can weight conclusions by vertical-level coverage.

7.7 Optional Floor Baseline

- **D4** – per-element-pair lookup table emitting the median descriptor value for the matching pair from the train bucket.
- Submitted as a floor, never as a baseline. Page-budget contingent.

8 Limitations and Broader Impact

9 Conclusion

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521 A Datasheet for Datasets

522 B License Terms

523 C Reproducibility Checklist

524 D Held-Out Construction Logic

525 D.1 H1 metal-ligand pair selection

- 526 • Element-pair co-occurrence (859 TM pairs in `metal_nonmetal_pairs.csv`) is a coarse
527 upper bound; selection here uses true bonds (Mayer / fuzzy bond order ≥ 0.3) extracted
528 from `bond.lmdb` via `tm_neighbor_lists.py`.
- 529 • Per-vertical bond extraction across 15 TM-bearing verticals; one row per (TM atom, partner
530 atom) bond.
- 531 • Global pair table: **1,101 unique (TM, partner) bond pairs** across **651,742 TM-bonded**
532 **structures** (16.35% of dataset).
- 533 • Frequency bands (log-spaced): B1 [1, 10) 529 pairs; B2 [10, 100) 93; B3 [100, 1k) 233; B4
534 [1k, 10k) 203; B5 [10k, 100k) 43.
- 535 • Stratified sampling shape (3, 2, 2, 4, 0), `numpy.random.default_rng(seed=87)`. B5
536 excluded (Pt-C, Ti-H, etc. dominate training); B4 capped at 4 to keep realized holdout near
537 15k.
- 538 • Selected pairs: *B1* Pt-Na, Rh-Fe, Ag-Pd; *B2* Sc-Ir, Cd-Sb; *B3* Mn-Sb, Ir-Ge; *B4* Tc-Cl,
539 Re-H, Sc-S, Hf-C.
- 540 • Variance: across 30 seeds with shape (3, 2, 2, 4, 0), realized size is $15,227 \pm 4,871$. Seed 87
541 lands within 200 of the 15k target.
- 542 • Reproducibility: `qtaim_gen/source/analysis/build_holdout_csvs.py`; outputs
543 `h1_metal_ligand_pair_definitions.csv` and `h1_metal_ligand_pairs.csv`.

544 D.2 H6 lanthanide-ligand pair selection

- 545 • Same construction as H1, applied to (Ln, partner) bonds via `tm_neighbor_lists.py`
546 `-element_class ln`.
- 547 • Five Ln-bearing verticals: `omol`, `tm_react`, `ml_mo`, `electrolytes_reactivity`,
548 `mo_hydrides`.
- 549 • Global pair table: **453 unique (Ln, partner) bond pairs** across **66,677 Ln-bonded struc-**
550 **tures** (1.67% of dataset). All 15 lanthanides represented.
- 551 • Frequency bands: B1=207, B2=102, B3=86, B4=58, B5=0 (max pair size 4,439). H6
552 effectively samples four bands.
- 553 • Stratified sampling shape (3, 2, 2, 1, 0), `seed=190`. B4 toned down to 1 vs H1's 4 (Ln B4 is
554 sparser, target $\sim 2.5k$).
- 555 • Selected pairs: *B1* Lu-Ni, Ce-La, Gd-Sb; *B2* Ho-F, Yb-Si; *B3* Dy-S, Nd-S; *B4* Tm-N.
- 556 • Reproducibility: same build script as H1; out-
557 puts `h6_lanthanide_ligand_pair_definitions.csv` and
558 `h6_lanthanide_ligand_pairs.csv`.

559 D.3 Holdout-Aware Splitting Pipeline

- 560 • **Step 1 – Pull holdouts.** `pull_holdout_records.py` reads each holdout
561 CSV, materializes per-set LMDBs under `data/holdouts/<set>/`, and emits a
562 `manifest_holdout.parquet` keyed on the LMDB key convention `relpath(folder,`
563 `root_dir).replace(os.sep, "__")`.

- **Step 2 – Exclude.** `multi_vertical.load_exclusion_set` re-derives LMDB keys from the manifest `rel_path` column and unions H1/H3/H6/H7/H8 into one per-vertical exclusion set.
- **Step 3 – Split.** `split_descriptor_lmdbs.py` reads `structure.lmdb`, drops excluded keys, assigns each remaining key to train/val/test by `BLAKE2b(formula_hill) % 10`, then partitions each of the 8 descriptor LMDBs (`structure`, `charge`, `qtaim`, `bond`, `fuzzy`, `other`, `orca`, `timings`) into the assigned split.
- Output layout: `<output_dir>/<vertical>/<split>/<descriptor>.lmdb`.
- HPC parallelism: `-include_verticals X + -merge_from DIR`, mirroring the pull pipeline. Per-vertical `split_assignment.json` is the durable, race-free record (no global plan file).
- Lock-step audit: across all 8 descriptors, `source row count = train + val + test + excluded` for every vertical (verified on `ani2x`: 215,642 source = 168,783 + 17,417 + 29,442, $n_{\text{excluded}} = 0$).

E Per-Method Descriptor Definitions

F HPC Configuration Files

G Per-Vertical Census Table

Table 3 expands Sec. 6 into the full per-vertical breakdown (34 verticals plus the corpus total). Figure 4 reports per-structure descriptor density for atoms and QTAIM bond critical points.

Table 3: Per-vertical census of OMol-Descriptors-4M. Counts are corpus-level pre-split totals. *Structs*: structure count. *Forms*: unique alphabetical formulas. *Atoms*: atom-level descriptor records. *Charge rec.*: per-job per-method charge-scheme records. *BCPs*: QTAIM bond critical points. *Mayer / Löwdin / Fuzzy*: per-scheme bond-order records.

Vertical	Structs	Forms	Atoms	Charge rec.	BCPs	Mayer	Löwdin	Fuzzy
5A_elytes	14,200	11,294	2,325,187	99,383	2,982,866	4,474,371	6,655,293	4,244,442
ani1xb	737,208	48,320	10,834,251	7,318,728	10,801,957	12,465,497	30,634,609	28,118,840
ani2x	215,642	5,773	3,280,277	1,586,066	3,346,555	3,824,490	9,516,073	6,379,670
dna	24,647	4,312	2,186,892	172,529	2,559,225	3,843,725	7,372,380	4,907,003
droplet	18,694	3,376	1,109,505	130,858	1,457,592	1,653,393	2,805,247	1,785,875
electrolytes_reactivity	66,967	18,063	874,449	468,769	887,548	986,445	2,340,553	1,953,147
electrolytes_redox	46,069	33,579	4,017,583	322,483	4,945,260	7,442,261	11,512,752	9,064,929
electrolytes_scaled_sep	30,768	21,186	2,307,396	215,369	2,847,853	11,683,800	8,801,619	5,116,940
geom_orca6	199,805	36,054	10,321,888	1,398,634	11,579,867	18,310,972	35,343,998	22,839,614
low_spin_23	12,939	12,045	927,730	90,573	1,113,772	2,194,086	3,228,506	2,213,493
ml_elytes	51,560	35,054	7,608,642	360,904	9,672,793	15,638,665	22,494,066	13,464,794
ml_mo	77,603	40,658	6,098,677	543,153	7,336,479	12,931,862	20,795,278	17,937,539
ml_protein_interface	61,856	35,118	8,314,895	432,979	9,480,078	16,584,221	26,101,394	15,494,518
mo_hydrides	3,396	2,870	232,942	23,772	280,311	664,784	851,813	781,762
nakb	5,606	1,188	651,552	50,642	777,349	1,297,268	2,280,331	1,551,414
noble_gas	1,915	136	66,993	13,651	84,800	43,394	116,497	87,580
noble_gas_compounds	18	12	189	126	179	326	560	518
omol	1,428,279	593,876	92,071,794	9,997,947	109,334,787	170,492,089	291,413,475	218,967,571
orbnet_denali	51,102	30,339	2,305,764	357,714	2,594,095	4,028,552	7,875,696	5,188,774
pdb_fragments_300K	93,334	68,881	11,255,165	653,329	13,208,426	21,291,730	36,314,068	22,292,731
pdb_fragments_400K	92,698	68,485	11,177,181	648,881	13,058,825	20,774,760	36,149,798	22,215,370
pdb_pockets_300K	16,870	15,453	3,245,155	118,069	3,915,033	8,616,479	10,415,619	6,407,044
pdb_pockets_400K	3,204	3,092	630,613	22,422	701,104	1,658,125	2,019,313	1,222,919
pmechdb	49,832	28,171	2,951,782	379,418	3,316,002	6,028,189	9,928,066	6,572,474
protein_core	35,406	14,980	4,197,118	247,840	4,436,976	8,229,618	13,219,684	7,738,794
protein_interface	44,760	16,127	4,298,048	313,320	4,938,560	7,052,373	13,235,662	8,007,220
rgd_uks	75,316	980	1,337,124	527,212	1,340,336	1,504,737	3,810,675	3,314,638
rmechdb	2,803	731	58,528	28,024	59,403	76,928	171,675	150,328
rna	34,434	5,168	3,661,659	241,036	4,108,650	7,166,774	12,646,859	8,583,065
rpmd	18,652	2,992	1,412,713	152,365	1,721,533	2,025,287	3,893,696	2,478,985
scaled_separations_exp	53,180	37,168	5,020,630	372,260	5,549,452	5,214,145	13,616,510	8,822,261
spice	44,447	19,603	1,553,948	311,129	1,686,049	2,413,029	4,992,500	3,292,358
tm_react	158,189	86,819	9,209,119	1,107,323	10,873,610	17,179,722	29,890,848	22,389,065
translx	213,731	171	2,974,386	1,496,117	2,977,381	3,345,716	8,459,481	7,799,805
TOTAL	3,985,130	1,302,074	218,519,775	30,203,025	253,974,706	401,137,813	688,904,594	491,385,480

Table 4: Per-vertical LMDB conversion-quality audit. Verticals with all zeros across the four failure columns are shown as “-”.

Vertical	<i>N</i>	QTAIM fail	ORCA fail	Other fail	Max drop
omol	1,429,192	17	896	0	1
ani1xbb_reparse	737,210	2	0	0	0
ani2x	215,642	-	-	-	-
trans1x	213,731	-	-	-	-
geom_orca6	199,805	-	-	-	-
tm_react	158,190	0	1	0	0
pdb_fragments_300K	93,334	0	0	0	1
pdb_fragments_400K	92,698	-	-	-	-
ml_mo	77,603	1	23	0	0
rgd_uks	75,316	-	-	-	-
electrolytes_reactivity	66,974	0	7	0	0
ml_protein_interface	61,859	0	3	0	0
scaled_separations_exp	53,219	2	37	0	0
ml_elytes	51,599	1	38	0	0
orbnet_denali	51,102	-	-	-	-
pmechdb	49,832	-	-	-	-
electrolytes_redox	46,081	0	12	0	0
protein_interface	44,972	0	212	0	0
spice	44,450	1	2	0	4
protein_core	35,406	-	-	-	-
rna	34,745	0	311	0	0
electrolytes_scaled_sep	30,769	0	1	0	0
dna	24,647	-	-	-	-
droplet	18,694	-	-	-	-
rpmd	18,652	0	0	0	1
pdb_pockets_300K	16,927	0	57	0	0
5A_elytes	14,202	0	2	0	2
low_spin_23	12,939	-	-	-	-
nakb	5,606	-	-	-	-
mo_hydrides	3,396	-	-	-	-
pdb_pockets_400K	3,210	1	5	0	0
rmechdb	2,803	-	-	-	-
noble_gas	1,915	-	-	-	-
noble_gas_compounds	18	-	-	-	-
Total	3,986,738	25	1607	0	4

Figure 5 shows per-method descriptor coverage across all 34 verticals. Rows are descriptor methods grouped by family; columns are verticals ordered by structure count (largest left). Universal methods (Hirshfeld, ADCH, CM5, Becke charges; Fuzzy bond orders; ORCA Mayer/Löwdin/SCF/gradient/dipole) reach $\geq 99\%$ on every vertical. QTAIM critical-point records achieve $\geq 99.9\%$ coverage.

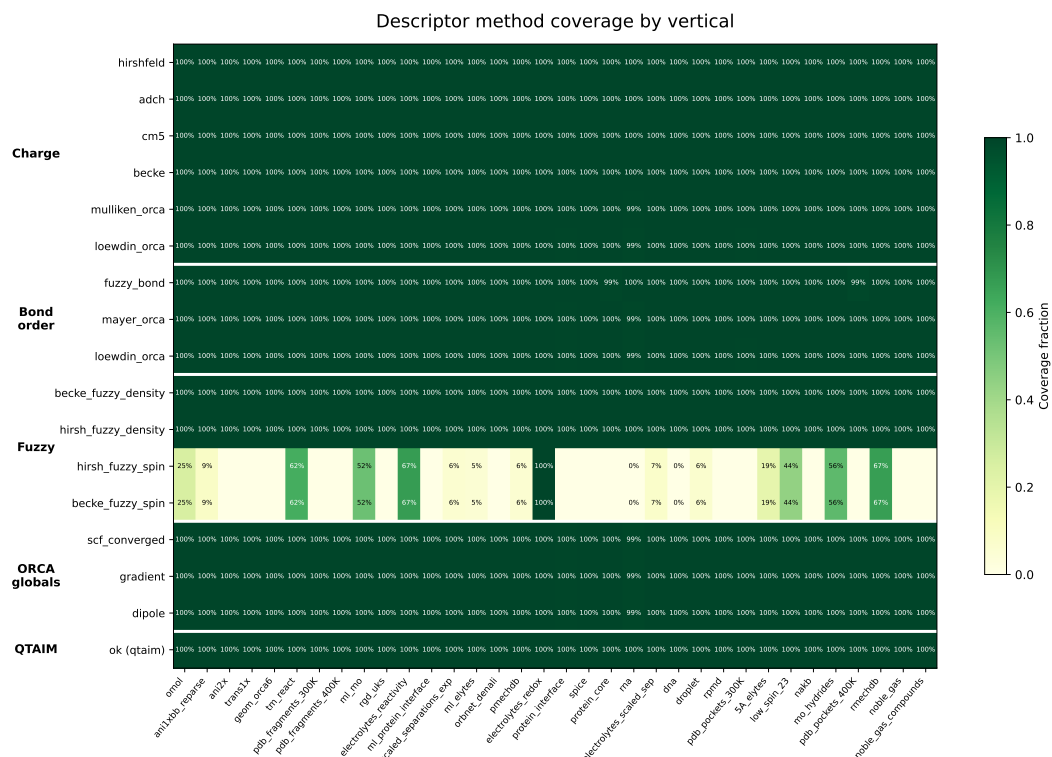


Figure 5: Per-method descriptor coverage fraction across all 34 verticals. Green = full coverage; white = absent. Vertical white lines separate descriptor families. The single ok (qtaim) column reports qtaim.lmdb record validity.

595 I Extended Bond-Order Histograms

596 J Additional Cross-Method Residual Plots

597 K Manifest Schema Documentation

598 L Code Structure Overview

599 M Throughput and Parallel Scaling

600 Here we report a local scaling experiment where we intent to show the concurrency performance of
 601 Multiwfn on our routines. The local sweep is run on 16 copies of a moderately-sized structure drawn
 602 from the dataset (from the droplet vertical here). It contains 36 atoms where we ran the three levels of
 603 calculations in triplicate with Multiwfn 3.8. We sweep the number of concurrent job-folder workers
 604 $\in \{1, 2, 4, 16\}$ at a fixed budget of cores (16) running to unearth how concurrency impacts wallclock.
 605 The configurations are W16T1 (16 workers $\times$ 1 thread/job), W4T4 (4 workers $\times$ 4 threads/job),
 606 W2T8 (2 workers $\times$ 8 threads/job), and W1T16 (1 worker $\times$ 16 threads/job). Here we see that
 607 single-threaded jobs yield the best performance with lower wallclock times across all three calculation
 608 levels. Thus, we recommend users to prioritize more workers with fewer threads per job assuming
 609 that you do not run into memory bottlenecks on larger systems with more memory-intensive routines.

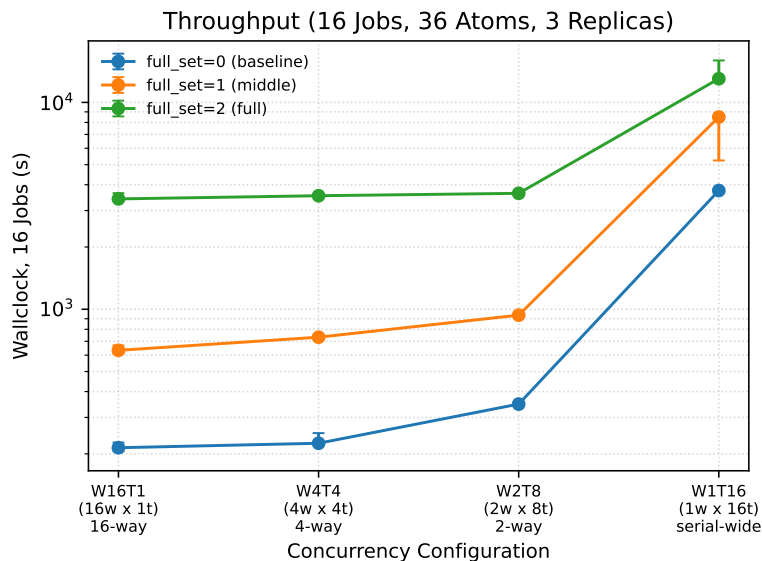


Figure 6: Multiwfn QTAIM post-processing wallclock as a function of intra-node concurrency configuration. Each configuration processes 16 homogeneous copies of `droplet_test` (36 atoms, def2-TZVPD) under cascade-restart tiers. Configurations saturate a 16-thread budget differently: W16T1 (16 workers $\times$ 1 thread), W4T4 (4 workers $\times$ 4 threads), W2T8 (2 workers $\times$ 8 threads), W1T16 (1 worker $\times$ 16 threads). Each curve represents a calculation tier (`full_set` 0=baseline, 1=middle, 2=full). Tiers 1 and 2 use `-restart` from the prior tier, so per-tier totals are cumulative phase wallclocks. Markers show medians across 3 replicates with min/max whisker error bars. Y-axis is logarithmic to visualize the $75\times$ range spanning W16T1 tier 0 (fastest) to W1T16 tier 2 (slowest). Hardware: AMD 7950X (16c/32t). ORCA 6.0.1 (`shared_openmpi416`) with system OpenMPI 4.1.2; Multiwfn 3.8 (Oct 2024 dev). ORCA SCF is precomputed; this measures post-processing only.